Three remaining structural tables over $\mathbb{F}_4$

This application verifies the subgroup descriptions in three current Connor–Leemans companion tables. The complete numerical inventories, source bijections and actual 44-point models are acquired from the accepted nine-table component. No ambient subgroup census is repeated.

Exact assertion. Relative to the acquired named atoms and structural registry, every one of the following 923 actual embedded subgroups admits its displayed source description. Colons assert split extensions, dots nonsplit extensions, and crosses direct products. These descriptions need not specify a unique abstract group; each occurrence has its own checked identification.

Current file	Displayed top group	Rows	Pages
`pgammal34`	$L_3(4) : S_3$	226	10
`psl34d2e2`	$L_3(4) : 2^2$	288	13
`psl34dD12`	$(L_3(4) : 2) : S_3$	409	19
Total		923	42

All forty-two native pages were directly inspected, including continued headers on pages without data. Powers, punctuation, explicit parentheses and default left association are retained. For example, the middle table has $2^{1+4} \cdot D_8$ in row 26, $2^{1+4} \cdot 4$ in row 44, and $Q_8 \cdot (2 \times 4)$ in row 85. The last table's top row is parsed as $(L_3(4) : 2) : S_3$, without replacing its displayed association by a filename shorthand. Its rows 146 and 147 both display $Q_8 \cdot D_8$ and require separate embedded joins.

The models are the acquired diagonal–field extension $P\Gamma L_3(4)$, the field–inverse-transpose extension with outer group C_2^2, and the full automorphism model, respectively. Their literal subgroup generators are parsed from their original saved states. No constructor or displayed group name can replace an embedded subgroup. The named $L_3(4)$ atom, including its convention for projective actions, is acquired from the previously accepted L100/M22 and atlas interfaces. Generic outer extension names do not assert a canonical numbering of automorphisms.

There are 272 acquired registry entries and no new named simple or Mathieu atom. Candidate generation proposes 47 new structural models: 27 split, 13 nonsplit and seven direct products. The new finite evidence consists of literal generator maps and factor/lift witnesses. Neither successful isomorphism search nor equality of group orders is accepted as a proof.

Acquired certificate arguments. A bijection fixing identity and preserving right multiplication by each ordered generator is a homomorphism, by induction on words. The reader checks all generator edges of every model-to-subgroup map. Factor names are acquired or processed earlier in an acyclic dependency order. For a split extension it verifies actual subgroups $N, K \leq H$, normality of N, $N \cap K = 1$, and $|N||K| = |H|$; hence $H = NK$. A noncommuting generator pair verifies the displayed nontrivial action. For a direct product the factor generators commute instead.

New cyclic-four and dihedral-eight lift tests. Let $N \triangleleft H$ be the actual checked normal subgroup. Suppose that $x \in H$, $x^4 \in N$, and the four cosets N, Nx, Nx^2, Nx^3 are pairwise disjoint and cover H. Then H/N is cyclic of order four. A complement exists if and only if some $a \in Nx$ satisfies $a^4 = 1$. Necessity follows by lifting the fixed quotient generator through the complement. Conversely such an a has order exactly four, since its quotient image does, and $\langle a \rangle \cap N = 1$, so it is a complement. Thus checking $a^4 \neq 1$ for every $a \in Nx$ certifies the printed dot.

For the dihedral test, take $x, y \in H$ satisfying

$$x^4,\ y^2,\ (xy)^2 \in N, \qquad H = \coprod_{j=0}^{1} \coprod_{i=0}^{3} Nx^i y^j.$$

The quotient is D_8: the presentation $\langle r, s \mid r^4 = s^2 = (rs)^2 = 1 \rangle$ reduces every word to $r^i s^j$, with $0 \leq i < 4$, $0 \leq j < 2$, while the eight disjoint cosets show these forms distinct. A complement exists if and only if there are $a \in Nx$, $b \in Ny$ with

$$a^4 = b^2 = (ab)^2 = 1.$$

A complement necessarily supplies these lifts. Conversely, such lifts define a homomorphism from the presented D_8 to H. Composing with $H \to H/N$ is the isomorphism sending r, s to Nx, Ny. Consequently the lift homomorphism is injective, its image meets N trivially, and its image maps onto H/N. It is the required complement. The certificate tests all $|N|^2$ pairs in the two specified cosets and requires each pair to fail at least one relation. It makes no inference from failure to discover a complement.

The reader first establishes group closure, the normal subgroup and its name, the quotient relations and every coset normal form. Thus neither quotient identity nor completeness of the lift choices is assumed. The three new cyclic-four and three new dihedral-eight certificates use these arguments. Four new prime-index models, one Klein-four quotient, one S_3 quotient and one $C_2 \times C_4$ quotient use the previously accepted complete lift tests, with their exact model data checked afresh.

Controls, source attribution and limits. Independent small permutation models test $C_8 \to C_4$ and $Q_{16} \to D_8$ as nonsplit examples. The split examples $C_2 \times C_4$ and $C_2 \times D_8$ must be rejected by the respective nonsplitting predicates. Damaged quotient generators, relations and trees test the boundary of each test. The existing controls also check actual source associations, powers, dots, degrees, embedded generators, literal-state grammar and terminal completion markers. Ordinary and optimized Python must produce identical complete reports.

One bounded witness pass returned all 923 proposals in 13.52 seconds, with clean output and no missing occurrence. The literal checker requires all source rows and all embedded joins. The current files credit Thomas Connor and Dimitri Leemans; `pgammal34` prints November 26, 2012, and the other two print November 29, 2012. Their present bytes are bound; identity with a particular historical 2013 or 2015 file is not asserted. No census, unique-extension classification, novelty, whole-original, question, formalization or measurement credit is claimed.